

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	0.4
Total 1 mark			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Uses sum of probs = 1	AO1.2	M1	$0.4 + b + c = 1$
Uses formula for $E(R)$	AO1.1a	M1	$b + c = 0.6$
Uses formula for variance $E(X^2) - (E(X))^2$	AO1.1a	M1	$E(R) = 0.2 \Rightarrow$ $(-2 \times 0.3) + (0 \times b) + (a \times c) + (4 \times 0.1) = 0.2$
Obtains a , b and c CAO	AO1.1b	A1	$ac = 0.4$ $E(X^2) - (E(X))^2 = (4 \times 0.3) + (0 \times b) + (a^2 \times c) + (16 \times 0.1) - (0.2)^2 = 3.56$ $a^2c = 0.8$ From (2) and (3) $a = 2$ Hence $c = 0.2$ and $b = 0.4$
Total 4 marks			

Q3.

(a) (i) $1 - \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right)$
OE

M1

$= \frac{1}{20} = 0.05$
AG

A1

2

(ii) $E(X) =$
 $1 \times \frac{1}{3} + 2 \times \frac{1}{4} + 3 \times \frac{1}{5} + 4 \times \frac{1}{6} + 5 \times \frac{1}{20}$
 At least 2 terms

M1

$$= 2.35$$

OE: give B2 for only 2.35 seen

A1

2

(iii) $E(X^2) =$
 $1 \times \frac{1}{3} + 4 \times \frac{1}{4} + 9 \times \frac{1}{5} + 16 \times \frac{1}{6} + 25 \times \frac{1}{20}$
 $(= 7.05)$

All 5 terms

$E(X^2) = 7.05$ with no working scores M0

Correct working but labelled $\text{Var}(X)$ and then no more done also scores M0

M1

$$\text{Var}(X) = E(X^2) - E(X)^2$$

Applied to this problem

$$= 1.5275$$

AG

m1

A1

3

(iv) $1 - (\frac{1}{3} + \frac{1}{4})$ or $(\frac{1}{5} + \frac{1}{6} + \frac{1}{20})$

M1

$$= \frac{5}{12} \text{ or } 0.417$$

©2025 Exam Papers Practice All rights reserved.

A1

2

(b) $'2.35' \times 100 - 50$

Their value of mean

M1

$$= 185$$

FT from (a)(ii)

Give B2 for only 185 seen

A1F

$$100^2 \times 1.5275 \text{ or } 100 \times \sqrt{1.5275}$$

M1

$$\text{SD} = \sqrt{15275} = 5\sqrt{611} = 124$$



A1

4

[13]

Q4.

(a)

n	Outcome	$P(N = n)$
1	H	0.5 ($\frac{1}{2}$)
2	TH	0.25 ($\frac{1}{4}$)
3	TTH	0.125 ($\frac{1}{8}$)
4	TTTH	0.0625 ($\frac{1}{16}$)
5	TTTTA	0.0625 ($\frac{1}{16}$)

B1 for **one** correct entry for $n = 1, 2, 4$

B2 for all 3 correct

Can be implied by correct $E(N)$

B2,1

$$E(N) = \left(1 \times \frac{1}{2}\right) + \left(2 \times \frac{1}{4}\right) + \left(3 \times \frac{1}{8}\right) + \left(4 \times \frac{1}{16}\right) + \left(5 \times \frac{1}{16}\right)$$

$$\sum_{n=1}^5 n \times P(N = n)$$

(all 5 terms attempted / seen / implied)

M1

$$= \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{16} = \frac{31}{16} = 1\frac{15}{16} \text{ (1.9375)}$$

(awfw 1.93 to 1.94)

A1

4

(b)

m	Outcome	$P(M = m)$
1	H	($\frac{1}{4}$)
2	TH	$\frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$
3	TTH	$\left(\frac{3}{4}\right)^2 \times \frac{1}{4} = \frac{9}{64}$
4	TTTH	$\left(\frac{3}{4}\right)^3 \times \frac{1}{4} = \frac{27}{256}$

(given)

(given)

(B1 any one correct)

(B2 any 2 correct)



5	TTTTA	$\left(\frac{3}{4}\right)^4 \times 1 = \frac{81}{256}$	(B3 all 3 correct)
---	-------	--	--------------------

B3,2,1

3

(c) (i) $P(J, R)$:

$$P(1,1) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8} \quad (\text{oe})$$

e.g 0.125

attempt at any $P(n,n)$

M1

$$P(2,2) = \frac{1}{4} \times \frac{3}{16} = \frac{3}{64} \quad (\text{oe})$$

$$P(3,3) = \frac{1}{8} \times \frac{9}{64} = \frac{9}{512} \quad (\text{oe})$$

$$P(4,4) = \frac{1}{16} \times \frac{27}{256} = \frac{27}{4096} \quad (\text{oe})$$

any 1 correct **to 3sf**

A1

$$P(5,5) = \frac{1}{16} \times \frac{81}{256} = \frac{81}{4096} \quad (\text{oe})$$

any 5 correct **to 3sf**

A1

$$p = \sum_{n=1}^{n=5} P(n,n)$$

©2025 Exam Papers Practice. All rights reserved.

$$\sum_{n=1}^{n=5} P(n,n) \text{ with all 5 values attempted}$$

m1

$$\Rightarrow p = \frac{221}{1024} \quad (0.2158)$$

(awfw 0.215 to 0.217)

A1

5

$$(ii) = 3 \times \left(\frac{221}{1024}\right)^2 \times \left(\frac{803}{1024}\right)$$

(either term with **their** p used) $(0 < p < 1)$

M1

$$+ \left(\frac{221}{1024} \right)^3$$

(Second term with **their** p used)
 $(0 < p < 1)$

M1

$$P(X \geq 2) = P(X = 2) + P(X = 3)$$

dep (M1M1)

Mdep1

$$= 0.120 \text{ (3dp)}$$

(allow 0.119; 0.12; 0.121)

A1

4

[16]

Q5.

(a)

r	1	2	3	4	5
p	.5	.24	.144	.0864	0.0296

$$0.4 \times 0.6 = 0.24$$

$$0.24 \times 0.6 = 0.144$$

$$0.144 \times 0.6 = 0.0864$$

B1 for any 1 correct (unsimplified) (B1)

B2 all correct and simplified

B2,1

2

(b) $P(\text{fewer than 3 bedrooms are rented})$
 $= P(R = 1, 2) \Rightarrow$

$$P(\text{fewer than 3 bedrooms not rented})$$

$$= 1 - P(R = 1, 2)$$

M1

$$= 1 - P(1 \text{ or } 2 \text{ rooms are rented})$$

$$= 1 - (0.5 + \mathbf{0.24})$$

[their $0 < p(2) \leq 0.4704$ value from table **used**]

$$= 1 - 0.74$$

m1

$$= 0.26$$

A1

Alternative:

P(fewer than 3 not rented)



$$= P(0, 1 \text{ or } 2 \text{ not rented})$$

$$= P(5, 4 \text{ or } 3 \text{ are rented})$$

$$= P(R = 3, 4, 5) \quad \text{M1}$$

$$p = 0.4 \times 0.6^2 + 0.4 \times 0.6^3 + 0.0296$$

$$= \mathbf{0.144 + 0.0864 + 0.0296} \quad \text{m1}$$

$$[\text{or their } p(3) + p(4) \leq 0.4704 \text{ value from table used}]$$

$$= 0.26 \text{ (cao)} \quad \text{A1}$$

$$[\text{SC 0.74 for B1}]$$

3

$$\begin{aligned} \text{(c) (i)} \quad E(R) &= 0.5 \times 1 + 0.4 \times 0.6 \times 2 + 0.4 \times 0.6^2 \times 3 + 0.4 \times 0.6^3 \times 4 + \\ &\quad 0.0296 \times 5 \\ &= 0.5 \times 1 + 0.24 \times 2 + 0.144 \times 3 + 0.0864 \times 4 + 0.0296 \times 5 \end{aligned}$$

$$\sum_{i=1}^5$$

$$\times P(R = r_i) \text{ from their table}$$

M1

$$= \mathbf{0.5 + 0.48 + 0.432 + 0.3456 + 0.148}$$

$$\left[= \frac{1}{2} + \frac{12}{25} + \frac{54}{125} + \frac{216}{625} + \frac{37}{250} \right]$$

$$(0.5 + 1.2576 + 0.148)$$

$$\therefore E(R) = 1.9056$$

$$[\text{awfw 1.9 to 1.91}] \left[1 \frac{566}{625} \right]$$

A1

2

$$\begin{aligned} \text{(ii)} \quad E(R^2) &= 0.5 \times 1^2 + 0.4 \times 0.6 \times 2^2 \\ &\quad + 0.4 \times 0.6^2 \times 3^2 + 0.4 \times 0.6^3 \times 4^2 + 0.0296 \times 5^2 \end{aligned}$$

$$[0.5 + 0.96 + 1.296 + 1.3824 + 0.74]$$

$$E(R^2) = 4.8784$$

AG

B1

$$\text{Var}(R) = 4.8784 - 1.9056^2 (= 1.24708864)$$

$$4.8784 - \text{their } E^2(R)$$

M1

$$= 1.25(\mathbf{3sf})$$

$$(\text{awfw 1.23 to 1.25})$$

A1

3

$$\text{(d)} \quad E(M) = 1250E(R) - 282 = 1250 \times 1.9056 - 282 = 2100$$

cao

B1

$$\text{Var}(M) = 1250^2 \times [4.8784 - 1.9056^2]$$

$1250^2 \times \text{their Var}(R) > 0 \text{ in (c)(ii)}$
(1 948 473 to 1 953 125)

$$\text{sd}(M) = 1250 \times \sqrt{1.24708864}$$

$$\text{sd}(M) = \sqrt{1948437} = 1395.9$$

$$(\sqrt{1953125} = 1397.5)$$

$$= 1395.91$$

(awfw 1395 to 1400)

A1

3

[13]

Q6.

(a) (i) $E(X) = \sum_x x \times P(X = x) = 28$

$$E(X^2) = \sum_x x^2 \times P(X = x) = 9$$

$$\text{Var}(X) = 9 - 2.8^2$$

(their $E(X^2)$ – their $E^2(X)$)

$$= 1.16$$

cao

B1

M1

©2025 Exam Papers Practice. All rights reserved.

A1

3

(ii) $E(S) = 3 \times E(X) = 8.4$
on their $E(X)$

B1ft

$$\text{Var}(S) = 3 \times \text{Var}(X) = 3.48$$

$3 \times \text{their Var}(X) \text{ from (i)} > 0$

NB There was a problem with part (a)(ii) which affected the marking of this part.

Please see the Report on the Examination for details.

B1ft

2

(b) $E(Y) = 3.5$

$$E(Y^2) = 13$$

for $E(Y)$

B1

$$\text{Var}(Y) = 13 - 3.5^2 = 0.75$$

on their $E(Y)$ and $E(Y^2)$

M1

$$\text{Var}(Y) > 0$$

A1ft

$$E(T) = 3 \times E(Y) = 10.5$$

cao

B1

$$\begin{aligned} \text{Var}(T) &= 3^2 \times \text{Var}(Y) \\ &= 9 \times 0.75 \\ &= 6.75 \end{aligned}$$

9 × their $\text{Var}(Y) > 0$
cao

M1

$$= 6.75$$

cao

A1

Alternative:

T	3	6	9	12	
T^2	9	36	81	144	
P	$\frac{1}{20}$	$\frac{2}{20}$	$\frac{3}{20}$	$\frac{14}{20}$	
$E(T)$	$\frac{3}{20}$	$\frac{12}{20}$	$\frac{27}{20}$	$\frac{168}{20}$	10.5
$E(T^2)$	$\frac{9}{20}$	$\frac{72}{20}$	$\frac{243}{20}$	$\frac{2016}{20}$	117

(M1A1)

(M1A1)

$$\text{Var}(T) = E(T^2) - [E(T)]^2$$

(used)

(M1)

$$\begin{aligned} &= 117 - 10.5^2 \\ &= 6.75 \end{aligned}$$

(A1)

6

(c) (i) $P(X > 1) = 0.8$

B1

1

(ii) $P(T = 3) = \frac{1}{20}$ and $P(T = 3 \text{ or } 6) = \frac{3}{20}$

$$P(X + T \leq 9 \text{ and } X > 1)$$

$$= P([2, 3 \text{ or } 6], [3, 3 \text{ or } 6], [4, 3])$$

$$= 0.1 \times \frac{3}{20} + 0.4 \times \frac{3}{20} + 0.3 \times \frac{1}{20}$$

$$= 0.015 + 0.06 + 0.015$$

$$= 0.09$$

Alternative:

$$P(T = 3) = \frac{1}{20} \text{ and } P(T = 6) = \frac{2}{20}$$

$$P(X + T \leq 9 \text{ and } X > 1)$$

$$= P([(2, 3, 4), 3], [(2, 3), 6])$$

$$= 0.8 \times \frac{1}{20} + 0.5 \times \frac{2}{20}$$

$$= 0.04 + 0.05 = 0.09$$

SC

B4

[any 4 correct p's from table B2]

[0.12 B3] [0.096 or 0.072 B2]

$$\begin{cases} P(T = 3) = \frac{1}{20} \text{ and } P(T = 3 \text{ or } 6) = \frac{3}{20} \\ P(T = 3) = \frac{1}{20} \text{ and } P(T = 6) = \frac{2}{20} \end{cases}$$

B1

4

(iii) $P(X + T \leq 9 \mid X > 1) = \frac{0.09}{0.80}$
 their (c)(ii)
 $\frac{0.09}{0.80} \quad (0 < p < 1)$

M1

$$= \frac{9}{80} (0.1125)$$

cao

A1

2

[18]

Q7.

(a) (i) $\alpha = \frac{25}{63}$ (OE)
 $\left(\frac{100}{252} \text{ or } \frac{50}{126} \text{ or } 0.397 \right)$

B1

1

(ii) $E(X) = 2.5$ (symmetry)

B1

1

(iii) $E(X^2) = \left(1 \times \frac{25}{252} \right) + \left(4 \times \frac{25}{63} \right) + \left(9 \times \frac{25}{63} \right) + \left(16 \times \frac{25}{252} \right) + \left(25 \times \frac{1}{252} \right)$
 $\sum x^2 \times p$ attempted

M1

$$E(X^2) = \frac{125}{18}$$

$$\left(6 \frac{17}{18} \text{ or } 6.94 \right)$$

©2025 Exam Papers Practice. All rights reserved. A1

$$\text{Var}(X) = \frac{125}{18} - \frac{25}{4}$$

$$[\text{their } E(X^2) - (\text{their } E(X))^2]$$

$$\text{dep } \sum x^2 \times p \text{ used}$$

m1

$$= \frac{25}{36}$$

$$0.694 \quad [\text{Var} > 0]$$

A1

$$\text{sd}(X) = \frac{5}{6}$$



$$0.833 \left(\sqrt{\text{their Var}(X)} \right) \text{ (dep m1)}$$

A1ft

5

(b) (i) $E(\text{Pay}) = \frac{4}{9} \times 90$ pence
 $= 40$ pence

M1

$\Rightarrow$ Joanne expected to make a loss
 (loss of 10p per game)

A1

Alternative:

$$\frac{5}{9} > \frac{2}{9} + \frac{2}{9} \Rightarrow \text{loss (for B1) then M1A1}$$

(ii) $E(\text{Loss}) = 100 \times 10$ pence
 $= £10$
 $100 \times (\text{their loss/game})$

B1ft

3

[10]

Q8.

(a) (i) $P(R \geq 5) = 0.3 + 0.25 + 0.1 + 0.05$
 $= 0.70$

CAO

B1

1

(ii) $E(R) = \sum rp$
 $= 3 \times 0.1 + 4 \times 0.2 + 5 \times 0.3 + 6 \times 0.25 + 7 \times 0.1 + 8 \times 0.05$

M1

$$= 5.2$$

A1

2

(iii) $E(R^2) = 9 \times 0.1 + 16 \times 0.2 + 25 \times 0.3 + 36 \times 0.25$
 $+ 49 \times 0.1 + 64 \times 0.05$
 (Correct expression **or** 28.7)

M1A1

$$(= 28.7)$$

$$\text{Var}(R) = 28.7 - 5.2^2$$

M1

$$= 1.66$$

AG

A1

4

(b) (i) $P(R + S = 6) = 0.1 \times 0.15 = 0.015$

$$P(R = 3 \text{ and } S \leq 5) = 0.1 \times 0.85 = 0.085$$

B1

$$P(R + S = 7) = 0.1 \times 0.4 + 0.2 \times 0.15$$

$$= 0.04 + 0.03$$

$$= 0.07$$

$$P(R = 4 \text{ and } S \leq 4) = 0.2 \times 0.55 = 0.110$$

B1

$$P(R + S = 8) = 0.2 \times 0.4 + 0.1 \times 0.3 + 0.3 \times 0.15$$

$$= 0.08 + 0.03 + 0.045$$

$$= 0.155$$

$$P(R = 5 \text{ and } S \leq 3) = 0.3 \times 0.15 = 0.045$$

B1

$$P(R + S \leq 8) = 0.015 + 0.07 + 0.155$$

$$P(R + S \leq 8) = 0.085 + 0.110 + 0.045$$

M1

$$= 0.24$$

$$= 0.24 \text{ (AG)}$$

©2025 Exam Papers Practice. All rights reserved.

A1

5

(ii) $p = {}^5C_4(0.24)^4(0.76) + (0.24)^5$

$$= 0.0126 + 0.000796$$

First term correct

+ correct second term

or *correct numerical values*

(must use $p = 0.24$)

M1M1

$$= 0.0134$$

$$(0.013 \text{ to } 0.0135)$$

A1

3

(iii) $P(R = 4 \mid R + S \leq 8)$

$$= \frac{P(R = 4 \text{ and } R + S \leq 8)}{P(R + S \leq 8)}$$

Alternative: (using (b)(i))

$$= \frac{P(R = 4 \text{ and } S \leq 4)}{P(R + S \leq 8)}$$

$$= \frac{0.11}{0.24} = \frac{11}{24}$$

$$= \frac{0.03 + 0.08}{0.24}$$

(numerator) **or** 0.11 seen

B1

(÷ 24 iff numerator < 0.24)

M1

$$= \frac{11}{24} (0.458)$$

CAO

A1

3

[18]

Q9.

(a)

r	1	2	3	4
$P(R = r)$	$\frac{2}{3}$	$\frac{2}{9}$	$\frac{2}{27}$	k

$$k + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} = 1 \Rightarrow k = \frac{1}{27}$$

M1

AG

A1

2

(b) $P(R \geq 3) = \frac{2}{27} + \frac{1}{27} = \frac{1}{9}$

Allow $\frac{3}{27}$ or 0.111

B1

1

(c) (i) $C = 27R + 5$

$$E(R) = \left(1 \times \frac{2}{3}\right) + \left(2 \times \frac{2}{9}\right) + \left(3 \times \frac{2}{27}\right) + \left(4 \times \frac{1}{27}\right)$$

$$= 1 \frac{13}{27}$$

$$(1.48) \text{ or } \frac{40}{27}$$

B1

$$\therefore E(C) = 27 \times 1 \frac{13}{27} + 5$$

M1

$$= 45$$

A1F

3

(ii)

$$E(R^2) = \left(1 \times \frac{2}{3}\right) + \left(4 \times \frac{2}{9}\right) + \left(9 \times \frac{2}{27}\right) + \left(16 \times \frac{1}{27}\right)$$

$$= 2 \frac{22}{27} \text{ or } \frac{76}{27}$$

$$(2.81)$$

B1

$$\text{Var}(R) = 2 \frac{22}{27} - \left(1 \frac{13}{27}\right)^2$$

©2025 Exam Papers Practice. All rights reserved.

M1

$$= \frac{452}{729}$$

$$(0.62)$$

$$\therefore \text{St. dev}^n(C) = 27 \times \sqrt{\frac{452}{729}}$$

$$27 \times \sqrt{\text{Var}(R)} [\text{Var}(R) > 0]$$

M1

$$= 21.3$$

$$\text{CAO } (21.26)$$

A1

4

$$\text{SC: } \text{Var}(C) = 452 \text{ (CAO)}$$

(B1M1B1A0)

Alternative (c)

C	32	59	86	113
P	$\frac{2}{3}$	$\frac{2}{9}$	$\frac{2}{27}$	$\frac{1}{27}$

C	32	59	86	113
n	18	6	2	1

$\bar{x} = 45$ and $\sigma = 21.260$

from calculator

$$\left(\bar{x} = \frac{\sum Cn}{27} \right)$$

[10]

Q10.

(a) (i) P(GG or YY or RR)

M1

$$= \frac{2}{10} \times \frac{1}{9} + \frac{3}{10} \times \frac{2}{9} + \frac{4}{10} \times \frac{3}{9}$$

$$= \frac{2}{9}$$

(AG)

A1

2

©2025 Exam Papers Practice. All rights reserved.

(ii)

$$P(\bar{B}\bar{B} \text{ or } \bar{B}B) = \frac{1}{10} \times \frac{9}{9} + \frac{9}{10} \times \frac{1}{9}$$

$$\frac{1}{10} + \frac{9}{10} \times \frac{1}{9}$$

M1

$$= \frac{1}{5}$$

(AG)

A1

2

(b) (i)

	Same	1 Blue	Neither
x	135	145	-45

$P(X = x)$	$\frac{2}{9}$	$\frac{1}{5}$	$\frac{26}{45}$
------------	---------------	---------------	-----------------

B1B1

2

$$(ii) \quad E(X) = 135 \times \frac{2}{9} + 145 \times \frac{1}{5} + (-45) \times \frac{26}{45}$$

Multiply two rows of their table from (b)(i)

M1

$$= 29 + 30 - 26 = 33 \text{ pence}$$

AG

A1

2

$$(c) \quad (i) \quad E(Y) = 104 - 3E(X)$$

$$= 104 - 3 \times 33 = 5 \text{ pence}$$

$\therefore$ Joanne would expect to win £5

OE (eg 500p)

M1

A1

A1

3

$$(ii) \quad E(X^2) = 9425$$

$$(4205 + 4050 + 1170)$$

©2025 Exam Papers Practice. All rights reserved.

B1

$$\text{Var}(X) = 9425 - 33^2 = \mathbf{8336}$$

$$sd(X) = 91.30$$

$$\text{Var}(Y) = 9 \times \text{Var}(X)$$

B1

$$= 9 \times 8336$$

$$9 \times (\text{their } \text{Var}(X) > 0)$$

$$= 75024$$

$$\text{or } 3 \times (\text{their } sd(X))$$

M1

$\Rightarrow$ standard deviation (Y) = 274 pence

273.9p or £2.74

A1

4

[15]

Q11.

(a) $P(X \geq 5) = P(X = 5) + P(X = 6)$
 $= \frac{5}{20} + \frac{6}{24}$

M1

$$= \frac{1}{2}$$

A1

2

(b) (i) $E\left(\frac{1}{X}\right) = \sum \frac{1}{x} \times P(X = x) =$
 $\left(1 \times \frac{1}{20}\right) + \left(\frac{1}{2} \times \frac{2}{20}\right) + \left(\frac{1}{3} \times \frac{3}{20}\right) + \left(\frac{1}{4} \times \frac{4}{20}\right) + \left(\frac{1}{5} \times \frac{5}{20}\right) + \left(\frac{1}{6} \times \frac{6}{24}\right)$
Use of $\sum \frac{1}{x} \times p$

M1

$$= \frac{1}{4} + \frac{1}{24}$$

$$= \frac{7}{24}$$

©2025 Exam Papers Practice. All rights reserved.

AG

A1

2

(ii) $E\left(\frac{1}{X^2}\right) = \frac{109}{900}$
Use of $\sum \frac{1}{x^2} \times p$

M1

or 0.21

A1

$$\text{Var}\left(\frac{1}{X}\right) = \frac{109}{900} - \left(\frac{7}{24}\right)^2$$

AG

A1

$$= 0.036 \text{ (3dp)}$$

3

(c) $A = \frac{1}{X}(X + 3)$

$$A = 1 + \frac{3}{X}$$

(either)

B1

$$E(A) = 1 + 3E(X^{-1})$$

$$E(A) = 1 + 3 \times \frac{7}{24}$$

(either)

M1

$$= 1 \frac{7}{8}$$

(1.875)

A1

$$\text{Var}(A) = \text{Var}\left(1 + \frac{3}{X}\right) = 9\text{Var}(X^{-1})$$

M1

$$= 9 \times \frac{173}{4800}$$

©2025 Exam Papers Practice. All rights reserved.

allow 9×0.036

$$= 0.324 \text{ or } \frac{519}{1600}$$

0.324375

A1

5

[12]

Q12.

(a) $\Sigma p = 1$

$$k = 1 - (0.40 + 0.25 + 0.18 + 0.12)$$

$$k = 0.05$$

B1

1

(b) (i) $E(X) = \sum_{\text{all } x} x P(X = x) = 3.17$

B1

1

(ii)
$$\text{Var}(X) = \sum_{\text{all } x} x^2 P(X = x) - \mu^2$$

$$\sum x^2 P(X = x) \text{ attempted}$$

M1

$$= 11.53 - 10.0489$$

$$\sum x^2 P(X = x) - [E(X)]^2$$

M1

$$= 1.4811$$

$$= 1.48$$

AWFW 1.48 - 1.49

A1

3

(c) (i)
$$E(Y) = 2E(X) - 3$$

$$= 2 \times 3.17 - 3$$

$$= 3.34$$

M1

1

(ii)
$$\text{Var}(Y) = 4 \text{Var}(X)$$

their Var(X) $\times 4$

M1

$$= 4 \times 1.48$$

$$= 5.92$$

 Standard deviation = 2.43

$\sqrt{\quad}$

Note: if separate table used for Y:

$$E(Y^2) = 17.08$$

$$[E(Y)]^2 = 11.16$$

m1A1

3

[9]

Q13.

(a) $\Sigma p = 1$

$$k = 1 - (0.01 + 0.05 + 0.14 + 0.30 + 0.12)$$

$$k = 0.38$$

B1

1

(b) (i) $E(X) = \sum_{\text{all } x} x P(X = x) = 4.35$

$$\frac{87}{20}$$

B1

1

(ii) $\text{Var}(X) = \sum_{\text{all } x} x^2 P(X = x) - \mu^2$

$E(X^2)$ attempted

M1

$$= 20.09 - 18.9225$$

$$\sum x^2 P(X = x) - \mu^2$$

M1

$$= 1.1675$$

$$\frac{467}{400} \text{ (AWFW 1.16 - 1.17)}$$

A1

3

(c) (i) $E(Y) = 5E(X) + 2$

$$= 5 \times 4.35 + 2$$

$$= 23.75$$

Their (b)(i) $\times 5 + 2$

M1

1

(ii) $\text{Var}(Y) = 25\text{Var}(X)$

$$= 29.1875$$

Their (b)(ii) $\times 25$

M1

Standard deviation = 5.40

$$\sqrt{\quad}$$

m1

$$(5.40 - 5.41)$$

A1

Q14.

(a) $1 - 3p \geq 0$ and $p \geq 0$

M1

$$3p \leq 1$$

A1

$$0 \leq p \leq \frac{1}{3}$$

A1

3

(b) (i) $E(X) = (1 \times p) + (2 \times p) + (3 \times p) + 4(1 - 3p)$
 $= 4 - 6p$

B1

1

(ii) $E(X^2) = (1 \times p) + (4 \times p) + (9 \times p) + 16(1 - 3p)$
 $= 16 - 34p$

Attempting $E(X^2)$

M1

$$\text{Var}(X) = (16 - 34p) - (4 - 6p)^2$$

M1

$$= 16 - 34p - 16 + 48p - 36p^2$$

$$= 14p - 36p^2$$

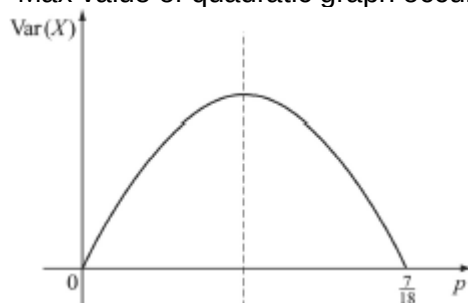
$$= 2p(7 - 18p)$$

AG

A1

3

(c) (i) Max value of quadratic graph occurs at:



Or

$$\frac{dV}{dp} = 14 - 17p \quad \text{and} \quad \frac{d^2V}{dp^2} = -17$$

$$\text{For max } V, \frac{dV}{dp} = 0 \quad \text{and} \quad \frac{d^2V}{dp^2} < 0$$

M1

$$p = \frac{1}{2} \left(0 + \frac{7}{17} \right) = \frac{7}{34}$$

$$\Rightarrow p = \frac{7}{34}$$

A1

2

(ii) Max value of $\text{Var}(X)$

$$= 2 \times \frac{7}{34} \times \left(7 - 17 \times \frac{7}{34} \right)$$

Attempted

M1

$$= 2 \times \frac{7}{34} \times \frac{7}{2}$$

$$= \frac{49}{17}$$

$$\text{sd}_{\max}(X) = \frac{7}{\sqrt{17}} (1.71)$$

(square root attempted)

M1

©2025 Exam Papers Practice. All rights reserved.

CAO

A1

3

[12]

Q15.

(a) (i)

x	4	-1
$P(X = x)$	$\frac{1}{5}$	$\frac{4}{5}$

B1

1

(ii) $E(X) = \left(4 \times \frac{1}{5} \right) + \left(-1 \times \frac{4}{5} \right) = 0$

$$(p > 0, \sum p = 1)$$

M1A1

2

(b)

x	4	-1
$P(X = x)$	$\frac{1}{3}$	$\frac{2}{3}$

B1

$$E(X) = \left(4 \times \frac{1}{3}\right) + \left(-1 \times \frac{2}{3}\right) = \frac{2}{3}$$

$$(p > 0, \sum p = 1)$$

B1

$$E(24X) = 24 \times E(X)$$

$$= 24 \times \frac{2}{3}$$

$$= 16$$

M1

A1

4

[7]

Q16.

$$\mu = E(X) = \sum_{i=1}^n x_i P(X = x_i)$$

(a)

$$\mu = (1 \times 0.09) + (2 \times 0.13) + (3 \times 0.50)$$

$$+ (4 \times 0.15) + (5 \times 0.12)$$

M1

$$\mu = 3.05$$

A1

$$\sigma^2 = \text{Var}(X) = E(X^2) - \mu^2$$

$$= 10.51 - 3.05^2$$

M1

$$= 1.2075$$

$$\therefore \sigma = \sqrt{1.2075} = 1.10$$

{1.099}

A1

4

(b) $P(\mu - \sigma < X < \mu + \sigma)$

$= P(1.95 < X < 4.15)$

M1

$= P(2 \leq X \leq 4)$

M1

$= 0.78$

A1

3

[7]

Q17.

(a) $\frac{1}{4}, \frac{1}{4}$ inserted in table

Accept statement, or use of correct probs in (b)

B1

1

(b) Method for mean

*Award M1 A0 if c then divides,
e.g. by 6*

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved. M1

Mean = 4

Allow NMS

A1

$E(X^2) = 19$

PI; allow even if c gives this as variance.

B1

Variance = $19 - 4^2 = 3$

*Alternative: $\sum p(x - 4)^2 = 3$ M1 A1
(-1 EE); Allow NMS*

M1 A1

[6]

Q18.

(a) $\frac{1}{6}$

allow 0.17

B1

1

(b) $1 \times \frac{1}{2} + 2 \times \frac{1}{3} + 3 \times \frac{1}{6} = \frac{5}{3}$ A.G

B1

4

$$E(X^2) = \frac{1}{2} + \frac{4}{3} + \frac{9}{6} \left(= \frac{10}{3} \right)$$

f.t. on part(a) seen

Variance = $\frac{5}{9}$ or 0.56

no working

0.56

$\frac{5}{9}$

0.55
SR2

SR1

3

M1A1f.t.

A1

[5]

Q19.

(a) (i) $E(X) = 0 \times 0.22 + 1 \times 0.25 + 2 \times 0.2 + 3 \times 0.14 + 4 \times 0.19$
 $= 1.83$

method

M1

1.83 cao

A1

(ii) $E(X^2) = 0 \times 0.22 + 1 \times 0.25 + 4 \times 0.2 + 9 \times 0.14 + 16 \times 0.19$
method

M1

$= 5.35$

(iii) $V(X) = 5.35 - 1.83^2 = 2.001$

*method for variance – may be calculated as $E[(X - E(X))^2]$
disallow if called standard deviation*

M1

$$\text{s.d.} = \frac{\sqrt{2.0011}}{\sqrt{\text{variance}}}$$

m1

$$= 1.41$$

$$1.41 - 1.42$$

$$\text{allow for variance} = 2.00 \text{ (} 2.00 - 2.01 \text{)}$$

A1

6

(b) (i) $P(X < 3) = 0.22 + 0.25 + 0.2 = 0.67$
0.67 cao

B1

(ii) $P(X > 1.83) = 0.2 + 0.14 + 0.19$
'their' mean
requires correct method for mean

m1

$$= 0.53$$

$$\text{0.53 cao}$$

A1

(iii) median = 2
2 cao

B1

$$P(X > 2) = 0.14 + 0.19 = 0.33$$

$$\text{0.33 cao}$$

B1

5

[11]

Examiner reports

Q3.

The majority of students scored very well on this question, although a few treated X as a continuous variable and performed various integrations and so scored few, if any, marks. Students should have appreciated that, in part (a)(iii), where they were required to 'Show that $\text{Var}(X) = 1.5275$ ', simply quoting values such as $E(X^2) = 7.05$ without any working, was not acceptable. This 'general principal' can be expected to apply on future papers. Part (b) was well done, although a small proportion of students, having found the variance, did not proceed to the standard deviation.

Q4.

Most candidates were successful in completing the given table in part (a). Unfortunately, some then failed to calculate the value of $E(N)$ as requested whilst others changed the 'given' values in Table 1, thus losing marks. In part (b), Table 2 was often completed but not always successfully. However, candidates in the main then realised that they needed to use the values found in Table 1 and Table 2 to help them to calculate the required probabilities in part (c)(i). This usually resulted in at least 3 marks being gained. Their result to part (c)(i) was then often correctly used in part (c)(ii), again with at least 3 marks being gained.

Q5.

This was the best answered question on the paper with many fully correct solutions seen. Almost all students managed to complete the table in part (a) correctly. Weaker students found part (b) very difficult to fathom. However, there were very many fully correct solutions with most making good use of the probabilities found in part (a). Part (c) was especially well done with many correct solutions to all parts. However, some students lost credit for using premature approximation or for failing to write down the value of the standard deviation having correctly found the value for the variance.

Q6.

©2025 Exam Papers Practice. All rights reserved.

In part (a)(i), candidates usually managed to find the correct values for $E(X)$ and $\text{Var}(X)$. Part (a)(ii) was considered to be invalid, as described at the beginning of this report. In part (b), there were many excellent solutions seen. Unfortunately, as is usual when the probability distribution is defined using formulae, several candidates thought the distribution to be continuous and consequently used integration methods in their attempts at $E(T)$ and $\text{Var}(T)$ for no reward.

The majority of candidates wrote down the correct value of 0.8 as their answer to part (c)(i). Part (c)(ii) was only answered well by the most able of candidates, and so the correct answer of 0.09 was very rarely seen. Many candidates attempted to find the correct probabilities with some success. However, they then multiplied their answer by 0.8 (or the value they had obtained in part (c)(i)), which rather negated the good work that they had done up to that point. Probably the most efficient ways of doing this part of the question were either to evaluate

$P(X = 2, 3 \text{ or } 4) \times P(T = 3 \text{ or } 6) - P(X = 4 \text{ and } T = 6)$ as $0.8 \times 0.15 - 0.01 \times 0.3 = 0.09$ or to complete a table of the required probabilities as shown below.

	$X = 2$	$X = 3$	$X = 4$
$T = 3$ (0.05)	$X + T = 5$ 0.05×0.1 $= 0.005$	$X + T = 6$ 0.05×0.4 $= 0.02$	$X + T = 7$ 0.05×0.3 $= 0.015$
$T = 6$ (0.1)	$X + T = 8$ 0.1×0.1 $= 0.01$	$X + T = 9$ 0.1×0.4 $= 0.04$	$X + T = 10$ 0.1×0.3 $= 0.03$

Some candidates included $X = 1$ and consequently gained an answer of 0.12, which some then again multiplied by 0.8 to obtain the very popular incorrect answer of 0.096. Only the most able candidates were able to answer part (c)(iii) correctly.

Q7.

The great majority of candidates gave the correct answer to part (a)(i), though a small minority stated that $a = 100$. In part (a)(ii), although most candidates gave the correct answer, not many were able to *write down* the value of $E(X)$ by looking at the symmetry of

the distribution; the majority of candidates used $E(X) = \sum_{\text{all } x} x \times P(X = x)$ to find the correct answer. Part (a)(iii) was done well with very many correct answers seen.

However, some candidates, having found that $\text{Var}(X) = \frac{25}{36}$, failed to go on to calculate the standard deviation. Some of those candidates, who used decimals rather than fractions, occasionally ended up with $\text{Var}(X) = 0.69$ giving $\text{sd}(X) = 0.831$ and as a consequence, lost an accuracy mark. It is emphasised that, as final answers are normally required to be given correct to 3 significant figures unless otherwise stated, working throughout should be to at least 4 significant figures.

Part (b) of this question was not well done by the majority of candidates. The most popular misconception, which gained no marks, was that Joanne made a loss because $P(\text{Loss}) = \frac{7}{9} > P(\text{Win}) = \frac{2}{9}$. In fact $P(\text{Loss}) = \frac{5}{9} > P(\text{Win}) = \frac{4}{9}$ which lead to the conclusion that Joanne could expect to make a loss.

When the correct reason for a loss was given, candidates were often then unable to

calculate the magnitude of the loss with $\frac{5}{9} \times 100 \times 10 - \frac{4}{9} \times 100 \times 90 = -1222 \frac{2}{9}$ often seen.

Candidates should have calculated $\frac{5}{9} \times 100 \times 50 - \frac{4}{9} \times 100 \times 40 = 1000\text{p}$ (£10) or

$50 \times 100 - \frac{4}{9} \times 100 \times 90 = 1000\text{p}$ (£10). A considerable number of candidates' answers

were penalised by 1 mark for rounding $\frac{5}{9} \times 100$ to 56 and $\frac{4}{9} \times 100$ to 44. This led to $56 \times 50 - 44 \times 40 = 1040\text{p}$ (£10.40) as the amount lost by Joanne.

Q8.

Part (a) was usually answered well. However, in part (a)(iii), too many candidates did not show a sufficiently complete and accurate method to gain full marks.

There were many correct solutions seen to part (b). In part (b)(i), those candidates who seemed to have very little or no idea simply attempted to multiply the corresponding values in the two tables together and then wrote the given answer of 0.24; this gained no credit.

In part (b)(ii), most candidates realised that binomial probabilities were required, but unfortunately candidates often only considered one term. Even when two terms were attempted, the incorrect expression $0.24^4 \times 0.76 + 0.24^5$ was often seen.

Part (b)(iii) was only answered correctly by the most able candidates. Some found $P(R = 4 \text{ and } S \leq 4)$ and thought that this was the required answer, whilst others realised that they had to divide $P(R = 4 \text{ and } R + S \leq 8)$ by 0.24 but were unable to find the correct value for the numerator.

Q9.

Far too many candidates treated the given distribution as *continuous* rather than *discrete* and consequently attempted to apply integration methods throughout. This obviously met with no success but also wasted a lot of time. Part (a) was usually done quite well. However, it must be pointed out that when an answer is given in the question, then an appropriate method must be shown by candidates in order to gain full credit. Simply evaluating $P(R = r)$ for $r = 1, 2, 3$ and then writing down the given answer was not

sufficient; some mention of $\sum_{\text{all } r} P(R = r) = 1$ had to be seen. $P(R = 4) =$

$\frac{2}{3} \left(\frac{1}{3} \right)^{4-1} = \frac{1}{27} = k$ was also seen but again gained no marks.

In part (b), some candidates incorrectly interpreted 'at least 3' as either meaning 'greater than 3' or 'less than or equal to 3' or even 'equal to 3', with $\frac{1}{27}, \frac{1}{26}$ and $\frac{2}{27}$ as common wrong answers.

There were very many correct solutions seen to parts (c)(i) and (c)(ii). The most common cause for loss of marks was in part (c)(ii), where a number of candidates correctly found $\text{Var}(C) = 452$ but then did not go on to attempt to find the standard deviation of C as requested. Several candidates confused the use of R and C .

Q10.

This was often the worst answered question on the paper. Many candidates were unable to show enough convincing working in part (a) to gain full credit. However, part (b) was usually done quite well, except by those candidates who simply wrote down a 2×2 table or stated +45 rather than -45. Most realised that the probabilities had to add up to one, so

correctly obtaining $\frac{26}{45}$.

The values in the table were then usually used correctly to obtain the given value of $E(X)$. Part (c)(i) was usually done quite well with many finding $E(Y) = 5$. Unfortunately some candidates failed to multiply this answer by 100 to give the final answer of 500 pence or £5. There were some excellent solutions, by various methods, to the final part of this question, but these were mainly given by the more able candidates.

Q11.

Although it was stated clearly in the question that X was a discrete random variable, many candidates employed integration methods throughout this question, all to no avail. Those who did treat the variable as discrete usually did very well, with many gaining full marks. Although there were many fully correct answers of 0.5 seen to part (a), there were several candidates who, having lost both marks here, went on to gain all other available marks for the question.

In part (b), where the answers were given, it was very pleasing to see candidates usually supplying the required valid working to enable them to gain full credit.

Unfortunately, there were still a few candidates who calculated $E(X) = \frac{17}{4}$ and then

wrongly assumed that $E\left(\frac{1}{X}\right) = \frac{1}{E(X)}$. A similar incorrect argument followed when

attempting to find the value of $\text{Var}\left(\frac{1}{X}\right)$.

Part (c) was very well done, even by some of those candidates who failed to gain any credit on the previous parts of this question.

Q12.

Candidates found this question very much to their liking, with the vast majority gaining full marks. Most candidates easily obtained 0.05 for the value of k in part (a), before using this value to help to obtain the correct answer of 3.17 for $E(X)$ in part (b)(i). Almost all

candidates managed to calculate $E(X^2) = \sum x^2 \times P(X = x)$ in part (b)(ii), with very few wrongly thinking that this was the same as the variance. In part (c), although the majority of candidates used the most efficient ways of finding $E(Y)$ and $\text{SD}(Y)$, by evaluating $E(Y) =$

$2 \times E(X) - 3$ and $\text{SD}(Y) = \sqrt{4 \times \text{Var}(X)} = 2\sqrt{\text{Var}(X)}$, there were frequent attempts made to tabulate the probability distribution for Y followed by the use of methods already employed in part (b). Although this process was often successful, it used up valuable time which could have been allocated to other more demanding parts of the paper.

Q13.

This proved to be a very popular question, with the overwhelming majority of candidates gaining full marks. Parts (a) and (b)(i) were done well, with very few numerical errors

seen. Almost all candidates managed to calculate $E(X^2) = \sum x^2 \times P(X = x)$ in part (b)(ii), with very few candidates wrongly thinking that this was the same as the variance.

In part (c), although the majority of candidates used the most efficient ways of finding $E(Y)$

and $\text{SD}(Y)$ by evaluating $E(Y) = 5 \times E(X) + 2$ and $\text{SD}(Y) = \sqrt{25 \times \text{Var}(X)} = 5 \times \sqrt{\text{Var}(X)}$, there were frequent attempts made to tabulate the probability distribution for Y , followed by the use of methods already employed in part (b). Although this process was often successful, it used up valuable time which could have been allocated to other more demanding parts of the paper. A few confused candidates thought that $\text{Var}(Y) = 25 \times E(X)$, whilst others simply found the variance, not realising that the standard deviation was requested.

Q14.

This was the worst-answered question on the paper with the vast majority of candidates gaining fewer than half marks. The fact that probability values must fall within the range $0 \leq p \leq 1$ implied that $0 \leq p < 1$ by inspection of the first three values in the table. Inspection

of the fourth value gave $1 - 3p \geq 0$, implying that $p \leq \frac{1}{3}$. This was not usually explained well by the weaker candidates. In part (b), $E(X)$ and $E(X)^2$ were usually attempted correctly with many fully correct expressions in terms of p seen. The majority of candidates then attempted to use the correct method by stating and then using $\text{Var}(X) = E(X^2) - (E(X))^2$. Unfortunately, a lack of good algebraic skills let some candidates down. They were either unable to cope with $(4 - 6p)^2$, where $16 - 36p^2$ was a common wrong expression, or they could not cope with the negative signs when attempting subtractions. Thus, it was hardly surprising that the correct answer was usually seen even though the working did not really lead to the correct expression given on the question paper for $\text{Var}(X)$.

Part (c)(i) asked for the value of p which gave a maximum value of $\text{Var}(X)$. Differentiation methods were employed by candidates when tackling this part. In such cases it should be

remembered that it is then expected that both $\frac{dV}{dp} = 0$ and $\frac{d^2V}{dp^2} < 0$ should be considered. This was not always seen. No candidate considered properties of a quadratic in order to

gain the correct answer of $p = \frac{7}{36}$. In part (c)(ii), in most cases where candidates managed to find a value for the variance, they rightly went on to attempt to deduce a value for the standard deviation.

Q15.

The majority of candidates found this question to be quite straightforward with many scoring full marks in part (a). A few mistakes were seen in part (b), usually from those candidates who confused the 'total mark' with the 'total number of questions' to then quote

an incorrect answer of $\frac{16}{24}$.

Q16.

Most candidates made a very good attempt at part (a), especially in finding the mean of the given distribution. The previously common errors of leaving the answer as σ^2 when σ is

asked for, or using $\text{Var}(X) = \sum x^2 p$ were not usually seen. In part (b), the majority of candidates could arrive at $P(1.95 < X < 4.15)$ but some were then unsure as to how to continue, not realising that this was equivalent to finding $P(2 < X < 4) = 0.78$.

Q17.

This question was very well answered, with many candidates gaining full marks.

A common error in finding the mean was to divide the correct answer by 6. In calculating the variance, a substantial minority of candidates squared the probabilities instead of, or as well as, the values of the random variable. Another common mistake was to subtract the mean, rather than the square of the mean.

Q18.

Most candidates showed a good grasp of the mean but variance was less well understood.

A common error in part (b) was to use $\sum xp(x)^2$, or $\sum (xpx)^2$, instead of $\sum x^2p(x)$.

Q19.

This question was well answered. A few candidates thought that $E(X^2)$ and the variance of X were identical. Others confused variance and standard deviation. A number correctly identified and evaluated the variance but did not complete the calculation of the standard deviation. Some candidates were unable to identify the median and so unable to make progress in part (b)(iii). A few candidates used a normal distribution in part (b).



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.